

A One-Dimensional Critical Holder Border Classification for RKHS Sandwiches

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Source and Target

The source is Max Schöpple and Ingo Steinwart, *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?*, arXiv:2312.14711 [1]. In Section 4.2, Theorem 12 proves a packing-number obstruction for RKHS sandwiches between Holder spaces. For bounded subsets of $\mathbb{R}^d$ with non-empty interior, the source concludes that no RKHS can satisfy

$$\text{Hol}^\alpha(X) \hookrightarrow H \hookrightarrow \text{Hol}^\beta(X)$$

when $\alpha - \beta < d/2$, and no bounded-kernel RKHS can satisfy

$$\text{Hol}^\alpha(X) \hookrightarrow H \hookrightarrow \ell_\infty(X)$$

when $\alpha < d/2$. The paper then says that the bounds are tight up to the border cases $\alpha - \beta = d/2$, respectively $\alpha = d/2$.

these preparations our result reads as follows.

Theorem 12. *Let (X, d) be a metric space and $0 < \beta \leq \alpha \leq 1$. If X is connected and there exists an RKHS H over X with*

$$\text{Höl}_d^\alpha(X) \hookrightarrow H \hookrightarrow \text{Höl}_d^\beta(X), \quad (12)$$

then there exists a constant $C > 0$ such that for all $0 < \delta < \min\{1, \text{diam } X\}$ we have

$$N_d^{\text{pack}}(X, \delta) \leq C\delta^{-2(\alpha-\beta)}. \quad (13)$$

Moreover, if there exists an RKHS H over X such that

$$\text{Höl}_d^\alpha(X) \hookrightarrow H \hookrightarrow \ell_\infty(X), \quad (14)$$

then there exists a constant $C > 0$, such that (13) holds for $\beta := 0$ and all $0 < \delta < \min\{1, \text{diam } X\}$.

Let us consider a bounded $X \subset \mathbb{R}^d$ with non-empty interior. By Theorem 12 and the packing number bound (8) we conclude that there is no RKHS H satisfying (12) if $\alpha - \beta < d/2$. Moreover, there is no RKHS H satisfying (14) if $\alpha < d/2$. Hence, (1) is infeasible for $d > 2$ within the family of Hölder spaces. We observe that the difference of the smoothness between the Hölder spaces in (12) needs to grow proportional to the underlying dimension in order to enable (1).

In Section 4.5, (26) will generalize Hölder spaces over X , allowing arbitrarily large non-integer smoothness. In Theorem 21 we will show that if $\alpha - \beta > d/2$, then there exists an intermediate RKHS H such that we have

$$\text{Höl}^\alpha(X) \hookrightarrow H \hookrightarrow \text{Höl}^\beta(X)$$

and in Theorem 22 we will show that for $\alpha > d/2$ there exists an intermediate RKHS H such that

$$\text{Höl}^\alpha(X) \hookrightarrow H \hookrightarrow C^0(X)$$

holds. Hence, the bounds derived in Theorem 12 are tight up to the border cases $\alpha - \beta = d/2$, respectively $\alpha = d/2$. For general metric spaces (X, d) however, we do not know any condition that implies the existence of an RKHS H satisfying (12) or (14).

The requirement that X is a *bounded* subset of $\mathbb{R}^d$ is crucial to obtain positive answers to (1), as the following

Partial Result

This packet gives a one-dimensional classification of the first-order Holder border cases left open by the source paper. Throughout, $\text{Hol}^s([0, 1])$ denotes the bounded s -Holder functions with norm

$$\|f\|_s = \max \left\{ \|f\|_\infty, \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^s} \right\}, \quad 0 < s \leq 1.$$

Theorem 1. *The one-dimensional critical first-order Holder border has the following behavior.*

1. *Let $0 < \beta < 1/2$ and $\alpha = \beta + 1/2$. Then the inclusion*

$$\text{Hol}^\alpha([0, 1]) \hookrightarrow \text{Hol}^\beta([0, 1])$$

is not cotype 2. Consequently, there is no reproducing kernel Hilbert space H on $[0, 1]$ such that

$$\text{Hol}^\alpha([0, 1]) \hookrightarrow H \hookrightarrow \text{Hol}^\beta([0, 1]).$$

2. *The inclusion*

$$\text{Hol}^{1/2}([0, 1]) \hookrightarrow \ell_\infty([0, 1])$$

is not cotype 2. Consequently, there is no bounded-kernel RKHS H on $[0, 1]$ such that

$$\text{Hol}^{1/2}([0, 1]) \hookrightarrow H \hookrightarrow \ell_\infty([0, 1]).$$

3. *The endpoint $\alpha = 1, \beta = 1/2$ is positive: the Sobolev space*

$$H^1([0, 1]) = \{f \in L^2([0, 1]) : f \text{ has weak derivative } f' \in L^2([0, 1])\}$$

with norm

$$\|f\|_{H^1}^2 = \|f\|_{L^2}^2 + \|f'\|_{L^2}^2$$

is a bounded-kernel RKHS on $[0, 1]$, and

$$\text{Hol}^1([0, 1]) \hookrightarrow H^1([0, 1]) \hookrightarrow \text{Hol}^{1/2}([0, 1]).$$

Thus the one-dimensional border $\alpha - \beta = 1/2, 0 \leq \beta \leq 1/2$, has a negative non-Lipschitz side and a positive Lipschitz endpoint. The remaining higher-dimensional endpoint, not settled here, is the bounded-kernel case $\text{Hol}^1([0, 1]^2) \hookrightarrow H \hookrightarrow \ell_\infty([0, 1]^2)$.

Idea of the Proof

The source's obstruction tests one metric scale at a time. At the critical exponent, one scale is exactly balanced: 2^j intervals of length 2^{-j} and bumps of target norm $2^{-j/2}$ contribute a bounded amount to the cotype lower bound. The missing information is that every dyadic scale contributes this same bounded amount. Taking the first m scales at once therefore makes the target ℓ_2 -sequence norm grow like $\sqrt{m}$.

The domain Rademacher norm stays uniformly bounded because, for $\alpha < 1$, signed dyadic tents of height $2^{-j\alpha}$ form uniformly bounded families in Hol^α . This forces the cotype-2 constants of the inclusion to be infinite. Schölpple and Steinwart show that an RKHS sandwich would make the inclusion 2-factorable, and hence cotype 2; this gives the contradiction.

The endpoint $\alpha = 1, \beta = 1/2$ escapes this obstruction for a real reason, not just a limitation of the tents. On an interval, every Lipschitz function is absolutely continuous with essentially bounded derivative, so it belongs to the Sobolev RKHS $H^1([0, 1])$. Conversely, Cauchy–Schwarz gives $H^1([0, 1]) \hookrightarrow \text{Hol}^{1/2}([0, 1])$. This produces an actual intermediate RKHS at the last one-dimensional border point.

Dyadic Tent Lemma

Let

$$\theta(t) = \max\{1 - |2t - 1|, 0\}, \quad t \in \mathbb{R}.$$

Thus θ is supported on $[0, 1]$, equals 1 at $1/2$, and is Lipschitz. For $j \geq 0$ and $0 \leq k < 2^j$, define

$$\theta_{j,k}(x) = 2^{-j\alpha} \theta(2^j x - k), \quad x \in [0, 1].$$

Lemma 1. *For every $0 < \alpha < 1$ there is a constant C_α such that for every finite set $A \subset \{(j, k) : j \geq 0, 0 \leq k < 2^j\}$ and every choice of signs $\varepsilon_{j,k} \in \{-1, 1\}$,*

$$\left\| \sum_{(j,k) \in A} \varepsilon_{j,k} \theta_{j,k} \right\|_{\text{Hol}^\alpha([0,1])} \leq C_\alpha.$$

Proof. For a fixed level j , set

$$g_j(x) = \sum_{k:(j,k) \in A} \varepsilon_{j,k} \theta_{j,k}(x).$$

The supports at level j have disjoint interiors and meet only where both neighboring tents vanish. Hence

$$\|g_j\|_\infty \leq 2^{-j\alpha}, \quad \text{Lip}(g_j) \leq C 2^{j(1-\alpha)}$$

with an absolute constant C . The sup norm of $\sum_j g_j$ is bounded by $\sum_j 2^{-j\alpha}$.

Let $x, y \in [0, 1]$, $x \neq y$, and put $h = |x - y|$. Choose $J \geq 0$ so that $2^{-(J+1)} < h \leq 2^{-J}$, with the evident modification when $h > 1/2$. For levels $j \leq J$, the Lipschitz estimate gives

$$\sum_{j \leq J} |g_j(x) - g_j(y)| \leq Ch \sum_{j \leq J} 2^{j(1-\alpha)} \leq C_\alpha h^\alpha.$$

For levels $j > J$, the sup-norm estimate gives

$$\sum_{j > J} |g_j(x) - g_j(y)| \leq 2 \sum_{j > J} 2^{-j\alpha} \leq C_\alpha h^\alpha.$$

Combining the two estimates proves the Holder seminorm bound. $\square$

Proof of the Negative Assertions

Let $0 < \beta < 1/2$ and $\alpha = \beta + 1/2$. For $m \geq 1$, consider the finite family

$$A_m = \{(j, k) : 0 \leq j < m, 0 \leq k < 2^j\}.$$

By Lemma 1,

$$\mathbb{E} \left\| \sum_{(j,k) \in A_m} \varepsilon_{j,k} \theta_{j,k} \right\|_{\text{Hol}^\alpha([0,1])} \leq C_\alpha,$$

where the $\varepsilon_{j,k}$ are independent Rademacher signs.

On the other hand, each $\theta_{j,k}$ has a large β -Holder norm. Indeed, if $c_{j,k} = (k + 1/2)2^{-j}$ is its midpoint and $e_{j,k} = k2^{-j}$ is the left endpoint, then

$$\frac{|\theta_{j,k}(c_{j,k}) - \theta_{j,k}(e_{j,k})|}{|c_{j,k} - e_{j,k}|^\beta} = \frac{2^{-j\alpha}}{(2^{-j-1})^\beta} = 2^\beta 2^{-j(\alpha-\beta)} = 2^\beta 2^{-j/2}.$$

Therefore

$$\left(\sum_{(j,k) \in A_m} \|\theta_{j,k}\|_{\text{Hol}^\beta([0,1])}^2 \right)^{1/2} \geq 2^\beta \left(\sum_{j=0}^{m-1} 2^j 2^{-j} \right)^{1/2} = 2^\beta \sqrt{m}.$$

The cotype-2 constants of the inclusion $\text{Hol}^\alpha([0,1]) \hookrightarrow \text{Hol}^\beta([0,1])$ are therefore at least $2^\beta C_\alpha^{-1} \sqrt{m}$ for every m , and hence are unbounded. The inclusion is not cotype 2.

Schöpple and Steinwart prove that if an RKHS H satisfies

$$E \hookrightarrow H \hookrightarrow F,$$

then the inclusion $E \hookrightarrow F$ is 2-factorable [1, Theorem 6]. They also record that every 2-factorable operator is cotype 2 [1, Lemma 5 and Corollary 8]. Thus the non-cotype result excludes an intermediate RKHS.

For the bounded-kernel endpoint, take $\alpha = 1/2$ and $F = \ell_\infty([0,1])$. The same family satisfies

$$\|\theta_{j,k}\|_{\ell_\infty([0,1])} = 2^{-j/2},$$

so

$$\left(\sum_{(j,k) \in A_m} \|\theta_{j,k}\|_{\ell_\infty([0,1])}^2 \right)^{1/2} = \sqrt{m}.$$

Lemma 1 again gives a uniformly bounded Rademacher norm in $\text{Hol}^{1/2}([0,1])$. Hence the inclusion $\text{Hol}^{1/2}([0,1]) \hookrightarrow \ell_\infty([0,1])$ is not cotype 2, and the same 2-factorization obstruction excludes every bounded-kernel RKHS sandwich.

Proof of the Positive Endpoint

Proposition 1. *The Sobolev space $H^1([0,1])$, equipped with the norm*

$$\|f\|_{H^1}^2 = \int_0^1 |f(t)|^2 dt + \int_0^1 |f'(t)|^2 dt,$$

is a bounded-kernel RKHS on $[0,1]$. Moreover,

$$\text{Hol}^1([0,1]) \hookrightarrow H^1([0,1]) \hookrightarrow \text{Hol}^{1/2}([0,1])$$

continuously.

Proof. We use the standard continuous representative of each H^1 -class. If $f \in \text{Hol}^1([0, 1])$, then f is Lipschitz, hence absolutely continuous, and $|f'| \leq \text{Lip}(f)$ almost everywhere. Therefore

$$\|f\|_{H^1} \leq \|f\|_{\infty} + \text{Lip}(f) \leq 2 \|f\|_{\text{Hol}^1([0,1])},$$

so $\text{Hol}^1([0, 1]) \hookrightarrow H^1([0, 1])$.

Conversely, let $f \in H^1([0, 1])$. For $x, y \in [0, 1]$, the fundamental theorem for Sobolev functions and Cauchy–Schwarz give

$$|f(x) - f(y)| \leq \int_{\min(x,y)}^{\max(x,y)} |f'(t)| dt \leq \|f'\|_{L^2} |x - y|^{1/2}.$$

Also, since the interval has measure 1, there is a point $a \in [0, 1]$ with $|f(a)| \leq \|f\|_{L^2}$. Hence

$$|f(x)| \leq |f(a)| + \int_0^1 |f'(t)| dt \leq \|f\|_{L^2} + \|f'\|_{L^2} \leq \sqrt{2} \|f\|_{H^1}.$$

Thus $H^1([0, 1]) \hookrightarrow \text{Hol}^{1/2}([0, 1])$ continuously.

The same sup-norm estimate shows that every point evaluation satisfies

$$|f(x)| \leq \sqrt{2} \|f\|_{H^1}, \quad x \in [0, 1].$$

Thus point evaluations are continuous, and $H^1([0, 1])$ is an RKHS. The uniform bound on the evaluation norms implies that its reproducing kernel is bounded on the diagonal, hence $H^1([0, 1])$ is a bounded-kernel RKHS. $\square$

Verification and Novelty Notes

The proof is deterministic and uses no numerical computation. The key local estimate is Lemma 1; it is included explicitly to avoid relying on a black-box wavelet characterization of Holder–Besov spaces.

This result is partial relative to the full scope of the source paper, but it does classify the first-order one-dimensional Holder border. It closes the non-Lipschitz one-dimensional border negatively, closes the one-dimensional bounded-kernel endpoint $\alpha = 1/2$ negatively, and closes the one-dimensional Lipschitz endpoint positively via $H^1([0, 1])$.

This packet does not itself settle the higher-dimensional bounded-kernel endpoint

$$\text{Hol}^1([0, 1]^2) \hookrightarrow H \hookrightarrow \ell_{\infty}([0, 1]^2),$$

but a subsequent lane-7 packet on the two-dimensional Lipschitz endpoint settles that case negatively. The higher-smoothness Besov/Triebel border cases discussed later in the source paper remain outside this packet.

Before promotion, the run indexes were searched for arXiv id 2312.14711, title keywords, RKHS, Holder, Hölder, border cases, alpha=d/2, and 2-factorable; no existing packet or attempt was found. Bounded web searches for the source title with border cases, for Holder spaces 2-factorable RKHS, and for $C^{1/2}$ RKHS bounded kernel found the source arXiv record but no separate answer to this border case. A later deep pass checked nearby RKHS regularity and Sobolev-RKHS literature. Those sources gave useful context, but the endpoint H^1 argument above is elementary and self-contained.

Human Review Recommendation

Review should focus on three points. First, check the uniform signed tent estimate in Lemma 1. Second, confirm that the source paper’s “2-factorable implies cotype 2” obstruction applies to the inclusions with the same Holder and bounded-kernel conventions used here. Third, verify the short Sobolev endpoint proof in Proposition 1; this is the piece that turns the previous negative packet into a one-dimensional classification.

References

- [1] Max Schölpple and Ingo Steinwart. *Which Spaces can be Embedded in Reproducing Kernel Hilbert Spaces?* arXiv:2312.14711, 2023. <https://arxiv.org/abs/2312.14711>.